Maple Chapter – Key Concepts & Commands (Clean Notes)

1. Basic Rules & Commands

- **Prompt Symbol:** Maple uses the symbol (>) as the command prompt.
- **Ending Commands:**
  - Use a semicolon (;) → executes and displays the result.
  - Use a colon (:) → executes but hides the result.

- **Case Sensitivity:** Maple is case-sensitive. **Example:** x ≠ X

  factor() must be written in lowercase (NOT Factor).
- **restart:** Resets all variables and unloads all packages.
- **evalf:** Converts symbolic values to numerical form. **Example:**

  evalf(Pi, 5); → gives π up to 5 decimal places.

2. Algebraic Commands

- **expand:** Expands expressions. **Example:** expand((x+1)*(x+2)); → x^2 + 3x + 2
- **factor:** Factorizes expressions. **Example:** factor(x^2 + 2x + 1); → (x + 1)^2
- **simplify:** Simplifies expressions. **Example:** simplify((x^4 - y^4)/(x^2 - y^2));

  → x^2 + y^2
- **subs:** Substitutes values. **Example:** subs(x=2, x^2 + x + 1); → 7
- **solve:** Solves equations algebraically. **Example:** solve(5x + xy^2 = 3, x);
- **System of equations:** solve({eq1, eq2}, {x, y});

3. Calculus Commands (HIGH PRIORITY)

- **limit:** Finds limits. **Example:** limit(f, x = a);
- **One-sided limits:** limit(f, x = a, right); limit(f, x = a, left);
- **diff:** First derivative. **Example:** diff(sin(x), x); → cos(x)
- **Higher derivatives:** Syntax uses $ for order. **Example:** diff(5xy, x$2);

  → second derivative w.r.t x
- **int (indefinite integral): Example:** int(sin(x), x); → -cos(x)
- **int (definite integral): Example:** int(sin(x), x = 0..Pi);
- **taylor:** Taylor series expansion. **Example:** taylor(f(x), x = a, n);

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4. Matrices & Vectors

- **Matrix (standard form):** Matrix([[row1], [row2]]) **Example:** Matrix([[1,2,3],[4,5,6]]);

  → 2×3 matrix
- **Identity Matrix:** Matrix(n, shape = identity);
- **Load package:** with(LinearAlgebra);
- **Transpose:** Transpose(M);
- **Inverse:** MatrixInverse(M);
- **Vector:** Vector([a, b, c]);

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5. Numerical Methods & Graphics

- **fsolve:** Finds numerical (decimal) solutions. Use when exact solution is not possible.
- **plot:** Plots functions. **Example:** plot(sin(x), x = 0..Pi);
- **minimize:** Finds minimum value in a range. **Example:** minimize(exp(x), x = 0..10); → 1

Top MCQ "Must-Knows"

- **Syntax Rule:** Every command must end with (;) or (:)
- **Derivative Symbol:** In diff(f, x$2), the $ indicates order of derivative.
- **Packages:** Commands like: with(LinearAlgebra); with(Student[Calculus1]);

  are used to load special functions.
- **Case Sensitivity Trap:** factor → correct Factor → usually WRONG in MCQs

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Maple Chapter – Predicted MCQs

1. In Maple, if you want to execute a command but do not want to display the result in the output, the command should end with:

   (a) Semicolon (;)

   (b) Colon (:)

   (c) Period (.)

   (d) Dollar sign ($)

2. Which command is used to completely clear the Maple internal memory and reset all assigned variables?

   (a) clear

   (b) reset

   (c) restart

   (d) delete

3. To calculate the 4th derivative of the function f(x)=sin(x) with respect to x, the correct syntax is:

   (a) diff(sin(x), x, 4)

   (b) diff(sin(x), x$4)

   (c) diff(sin(x), x^4)

(d) Diff(sin(x), 4x)

4. To create a 2×2 Identity Matrix in Maple, which command is most efficient?

   (a) Matrix(2, shape=identity)

   (b) Matrix([[1,0],[0,1]])

   (c) Matrix(2, 2, identity)

   (d) Both (a) and (b) are correct

5. The command evalf(sqrt(2), 10) will result in:

   (a) The symbol √2

   (b) The number 2

   (c) The numerical value of √2 up to 10 decimal digits

   (d) An error message

6. Which Maple command is used to obtain a numerical (floating-point) solution for an equation where an exact algebraic solution is not possible?

   (a) solve

   (b) fsolve

   (c) nsolve

   (d) simplify

7. To find the limit of 1/x as x approaches 0 from the right side, the command is:

   (a) limit(1/x, x=0, right)

   (b) limit(1/x, x=0, +)

   (c) limit(1/x, x->0, right)

   (d) Limit(1/x, x=0)

8. Maple is a "case-sensitive" software; this means that:

   (a) X and x are the same variable

   (b) X and x are two different variables

(c) All commands must be in Uppercase

(d) All commands must be in Lowercase

9. The command expand((x + 5)^2) will output:

   (a) (x+5)^2

   (b) x^2 + 10x + 25

   (c) x^2 + 25

   (d) 2(x+5)

10. To substitute the value x=3 into the expression x^2 + 2x + 1, we use:

    (a) sub(x=3, x^2 + 2x + 1)

    (b) subs(x=3, x^2 + 2x + 1)

    (c) eval(x=3, x^2 + 2x + 1)

    (d) replace(x=3, x^2 + 2x + 1)

11. The correct format to load the Linear Algebra package for matrix operations is:

    (a) load(LinearAlgebra)

    (b) using(LinearAlgebra)

    (c) with(LinearAlgebra)

    (d) package(LinearAlgebra)

12. To plot the function cos(x) for values of x from $-\pi$ to π, the correct command is:

    (a) plot(cos(x), x = -Pi .. Pi)

    (b) plot(cos(x), -Pi .. Pi)

    (c) Plot(cos(x), x = -Pi .. Pi)

    (d) graph(cos(x), -Pi, Pi)

13. The output of the command factor(x^2 - 9) will be:

    (a) (x - 3)(x + 3)

(b) (x - 3)^2

(c) (x + 3)^2

(d) x^2 - 9

14. In Maple worksheet mode, the command prompt is visually represented by the symbol:

(a) $

(b) #

(c) >

(d) >>

15. To find the definite integral $\int_0^1 x^2\, dx$, the correct command is:

(a) int(x^2, x)

(b) int(x^2, x = 0..1)

(c) Int(x^2, 0..1)

(d) value(x^2, 0, 1)